



* April 5, 2001

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Mattia Garatti

Curriculum Vitae

I am a Ph.D. student in Mathematics interested in Mathematical Analysis, and its interaction with Mathematical Physics and Geometry. In my master thesis I have worked on the variational representations of optimal transport and Schrödinger problem: the aim has been to highlight how the first can be recovered as Γ -limit of the second. Currently, I am carrying out my research on regularized optimal transport problem(s).

Education

Ph.D. in Mathematics, *Université Paris-Saclay*, Orsay, France. **2025 – ongoing**

Thesis: "On regularized Optimal Transport problems". Supervisors: Luca Nenna - Simona Rota-Nodari.

Master degree in Mathematics (LM-40), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2023 – 2025**

Average: 30 cum laude. Main courses' topics: Real and Functional Analysis, Measure Theory, Differential Geometry, Mathematical Physics. Thesis: "Variational representations of optimal transport and Schrödinger problem". Supervisor: Luca Tamanini - Cosupervisor: Nicolò De Ponti.

Bachelor degree in Mathematics (L-35), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2020 – 2023**

Average: 31.33/30. Thesis: "Il Teorema di esistenza di Peano per le equazioni differenziali ordinarie", provision of a theory on ODEs in the Peano's assumptions. Supervisor: Marco Marzocchi

High School Diploma, *Liceo Scientifico 'Camillo Golgi'*, Breno (BS), **2015 – 2020**
100/100.

Talks

Variational representations of optimal transport and Schrödinger problem, **2025**
Math PhD Days at UniTO, Turin.

Talk on my master thesis work

Teaching

MathExcellence 2025: corso di eccellenza in Matematica, *Università Cattolica del Sacro Cuore*, Brescia (BS). **June 2025**

Assistance and tutoring activities (5 hours)

Teacher Assistant, internship at *Liceo Scientifico 'Golgi'*, Breno (BS). **September 2022**

Frontal lectures for high school students, exam administration and correction, laboratory assistant

Private Teacher, internship at *Cooperativa 'Assolo'*, Berzo Inferiore (BS). **August – September 2018**

Lessons of Mathematics and Physics for high school students with SLD

Teacher Assistant, *supervision of preschool children, scientific experiments*, **June 2018**
Fondazione Scuola Materna 'San Giuseppe', Gratacasolo (BS).

Work experience

Math writer, Zanichelli editore S.p.A., Bologna (BO).

2024 – 2025

Teamwork with the editorial staff, autonomous creation of exercises of various difficulty on different topics to be included in future publications

Summer schools and workshop attended

Modelling Viscoelastic Fluids, *Università Cattolica del Sacro Cuore*, Brescia. **February 2026**

Lecture series about mathematical models for viscoelastic fluids

Calculus of Variations and applications, *CIRM*, Marseille, France. **January 2026**

Workshop on the most recent advances in the field

Geometry, duality and convexity in new OT problems, *IMO Paris–Saclay*, Orsay, France. **October 2025**

Workshop on new Optimal Transport problems arising in various applications

PGMO Days 2025, *EDF Lab Paris–Saclay*, Palaiseau, France. **October 2025**

Workshop on Optimal Trasport and its interactions with Data Sciences

Advances in Calculus of Variations and Nonlinear Analysis, *Università Cattolica del Sacro Cuore*, Brescia. **September 2025**

Workshop in honor of Marco Degiovanni's 70th birthday

Il Problema di Plateau, *Università Cattolica del Sacro Cuore*, Brescia. **April 2025**

Lecture series about different approach to Plateau problem(s)

Introduction to Mechanics of Swelling, *Università Cattolica del Sacro Cuore*, Brescia. **February 2025**

Lecture series about mathematical models for swelling

Introduzione ai modelli matematici per i cristalli liquidi, *Università Cattolica del Sacro Cuore*, Brescia. **February 2024**

Lecture series about mathematical models for liquid crystals

Interuniversity Mathematics Summer School, *University of Perugia*. **July – August 2023**

Complex Analysis and PDE courses with related final exams successfully passed

Mathematics in Ljubljana, *University of Ljubljana, FMF and IMFM*. **July 2023**

Summer School with courses on Formalization of Mathematics and Introduction to Complex Dinamics

Storia e tecniche dei metodi geometrici di rappresentazione: un'introduzione, *Università Cattolica del Sacro Cuore*, Brescia. **March 2023**

Mini-course about cartography and perspective

Un'Introduzione alla Teoria dei Grafi, *Università Cattolica del Sacro Cuore*, Brescia. **February 2023**

Mini-course about Graph Theory and its applications

Alcuni Aspetti della Matematica Antica, *Università Cattolica del Sacro Cuore*, Brescia. **March 2021**

Webinar about history of Mathematics and some greek geometry aspects

Corso di Eccellenza in Matematica, *Università Cattolica del Sacro Cuore*, Brescia. **June 2019**

Mini-course for high school students about different fields of Mathematics

Computer skills

Programming languages: Phyton, C, C++

Mark-up languages: LaTeX, HTML

Languages

Italian: Mother tongue
English: Fluent
French: Pre-intermediate

Awards

"Smart" study award, *Università Cattolica del Sacro Cuore*, Brescia (BS). **2025**
Award for 100 best students of the university

Merit scholarship, *Tenaris*, Dalmine (BG). **2017 – 2024**
Scholarships awarded on merit every year during high school and university

Merit scholarship, *Consorzio del Bacino Imbrifero Montano dell'Oglio*, Bergamo. **2021 – 2025**
Scholarships awarded on merit every year during university

Exemption from first year university fees, *Università Cattolica del Sacro Cuore*, Brescia. **2020**
Exemption by merit

Merit scholarship, *Comune di Costa Volpino* (BG). **2020**
Scholarships awarded on merit for high school diploma

Organization

Member of the student jury of Galileo Prize. **2024**
Jury for award regarding scientific dissemination

Member of Joint Teachers – Students Commission, *Università Cattolica del Sacro Cuore*, Brescia. **2024 – 2025**

Staff member of Celebration of Mind, Notte dei Ricercatori, Venite a vedere la Terra che gira.
Mathematical popularization days, *Università Cattolica del Sacro Cuore*

Staff member of Disfida matematica. **2023 – ongoing**
Mathematical team competition for high school students, local phase in Brescia

Member of Ufficio di Sezione Elettorale, *Comune di Costa Volpino* (BG). **2021 – 2025**
Scrutineer and president of polling station

Parish volunteer, *Parrocchia di "San Zenone"*, Gratacasolo. **2015 – 2025**
Organization of summer camps and member of the local choir with budget manager functions

I hereby authorize the use of my personal data in accordance to the GDPR 679/16 - "European regulation on the protection of personal data".

February 07th, 2026



Mattia Garatti